

BISBEE DOUGLAS, AZ, USA

WMO: 722720

Lat: 31.458N

Lon: 109.606W

Elev: 1251

StdP: 87.17

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 93026

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.6	-3.8	-20.0	0.7	11.4	-17.7	0.9	11.8	12.5	14.1	11.1	12.9	1.7	0	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	18.2	37.7	16.8	36.4	16.7	35.1	16.7	20.8	28.4	20.5	28.0	20.1	27.5	5.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	16.4	22.1	18.8	15.9	22.0	18.1	15.2	22.1	66.3	28.3	64.5	27.8	62.9	27.2	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.6	8.4	-10.0	40.1	2.5	1.0	-11.8	40.8	-13.2	41.4	-14.6	42.0	-16.5	42.8		
			-11.2	22.3	2.4	0.7	-12.9	22.8	-14.3	23.1	-15.7	23.5	-17.5	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	8.0	9.9	13.1	16.5	21.1	26.3	26.8	25.8	23.7	18.3	12.4	7.8
	DBStd	7.54	3.36	3.48	3.37	3.41	3.06	2.45	2.01	1.88	2.14	3.40	3.53	3.35
	HDD10.0	235	78	40	13	2	0	0	0	0	0	1	18	83
	HDD18.3	1351	321	236	162	73	11	0	0	0	1	42	180	326
	CDD10.0	2978	16	38	109	196	343	489	522	489	412	258	88	17
	CDD18.3	1053	0	0	1	17	96	239	263	231	163	42	1	0
	CDH23.3	14037	4	30	177	581	1742	3543	2921	2338	1827	768	104	3
	CDH26.7	7053	0	2	23	160	824	2131	1605	1177	858	263	11	0
Wind	WSAvg	3.5	3.0	3.5	3.9	4.4	4.2	3.9	3.4	2.9	3.0	3.2	3.0	3.0
Precipitation	PrecAvg	316	19	14	12	4	5	10	79	77	33	25	14	24
	PrecMax	465	81	51	55	37	27	62	159	174	99	135	44	92
	PrecMin	170	0	0	0	0	0	0	27	4	0	0	0	0
	PrecStd	85	19	14	13	7	7	14	38	39	26	36	13	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	26.3	28.9	32.1	36.7	39.7	39.0	37.2	35.5	32.9	28.1	23.4
		MCWB	9.5	10.1	11.1	12.2	15.0	16.6	17.8	18.6	17.3	15.8	11.9	9.8
	2%	DB	21.3	23.9	27.1	30.0	34.4	38.2	37.4	35.6	34.1	31.2	26.3	21.3
		MCWB	8.5	9.2	10.5	11.8	13.9	16.2	17.8	18.7	17.2	14.9	11.5	9.1
	5%	DB	19.5	22.0	25.4	28.5	32.8	37.0	35.9	34.1	32.9	29.7	24.2	18.9
		MCWB	7.8	8.5	9.9	11.2	13.4	15.9	17.8	18.6	17.0	14.4	11.0	8.1
	10%	DB	17.4	19.7	23.5	27.0	31.3	35.6	34.2	32.6	31.6	27.9	22.3	16.8
		MCWB	7.1	7.8	9.3	10.7	12.8	15.6	18.0	18.9	17.0	13.8	10.3	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.8	12.7	14.0	16.5	20.0	21.6	21.8	20.5	18.4	14.1	11.8
		MCDB	17.5	19.1	24.3	28.3	31.0	29.2	28.9	29.6	27.6	26.7	20.2	17.9
	2%	WB	10.1	10.6	11.5	12.9	15.4	18.9	20.8	21.0	19.9	17.1	13.0	10.4
		MCDB	17.2	19.1	23.4	27.1	30.2	29.1	28.0	28.9	27.3	26.0	22.0	17.7
	5%	WB	9.0	9.7	10.6	11.9	14.5	18.0	20.4	20.6	19.3	16.1	11.9	9.3
		MCDB	16.8	19.0	22.8	26.4	29.7	29.9	27.6	28.4	26.8	25.4	21.5	16.7
	10%	WB	7.9	8.7	9.9	11.1	13.5	17.3	20.0	20.2	18.8	15.2	11.0	8.2
		MCDB	16.0	18.3	22.0	25.5	29.2	30.6	27.3	27.8	26.4	24.3	20.6	15.6
Mean Daily Temperature Range	5% DB	MDBR	17.9	17.9	18.5	18.9	19.0	18.2	14.1	13.9	15.3	17.9	18.4	17.2
		MCDBR	21.6	22.7	21.9	21.2	20.9	19.7	17.0	15.9	17.5	19.9	21.3	20.9
		MCWBR	11.8	11.5	10.4	9.8	9.2	7.7	5.0	4.2	5.9	8.6	10.7	11.3
	5% WB	MCDBR	17.6	18.1	19.3	19.1	18.8	16.1	13.2	13.4	13.5	15.8	17.7	17.2
		MCWBR	10.1	9.2	9.2	9.0	8.3	5.8	3.5	3.8	4.5	6.8	9.1	9.7
Clear-Sky Solar Irradiance	taub	0.250	0.252	0.271	0.282	0.300	0.327	0.376	0.359	0.328	0.285	0.259	0.248	
	taud	2.669	2.663	2.574	2.522	2.512	2.444	2.435	2.460	2.527	2.609	2.647	2.661	
	Ebn at Noon	1004	1027	1019	1009	985	952	905	919	942	972	982	988	
	Edh at Noon	73	81	95	105	107	114	115	111	99	85	74	70	
All-Sky Solar Radiation	RadAvg	3.72	4.68	6.14	7.47	8.18	7.93	6.58	6.30	5.82	5.10	4.08	3.43	
	RadStd	0.26	0.37	0.35	0.28	0.31	0.43	0.34	0.23	0.26	0.31	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-77	+75	N/A

Nomenclature: See separate page